

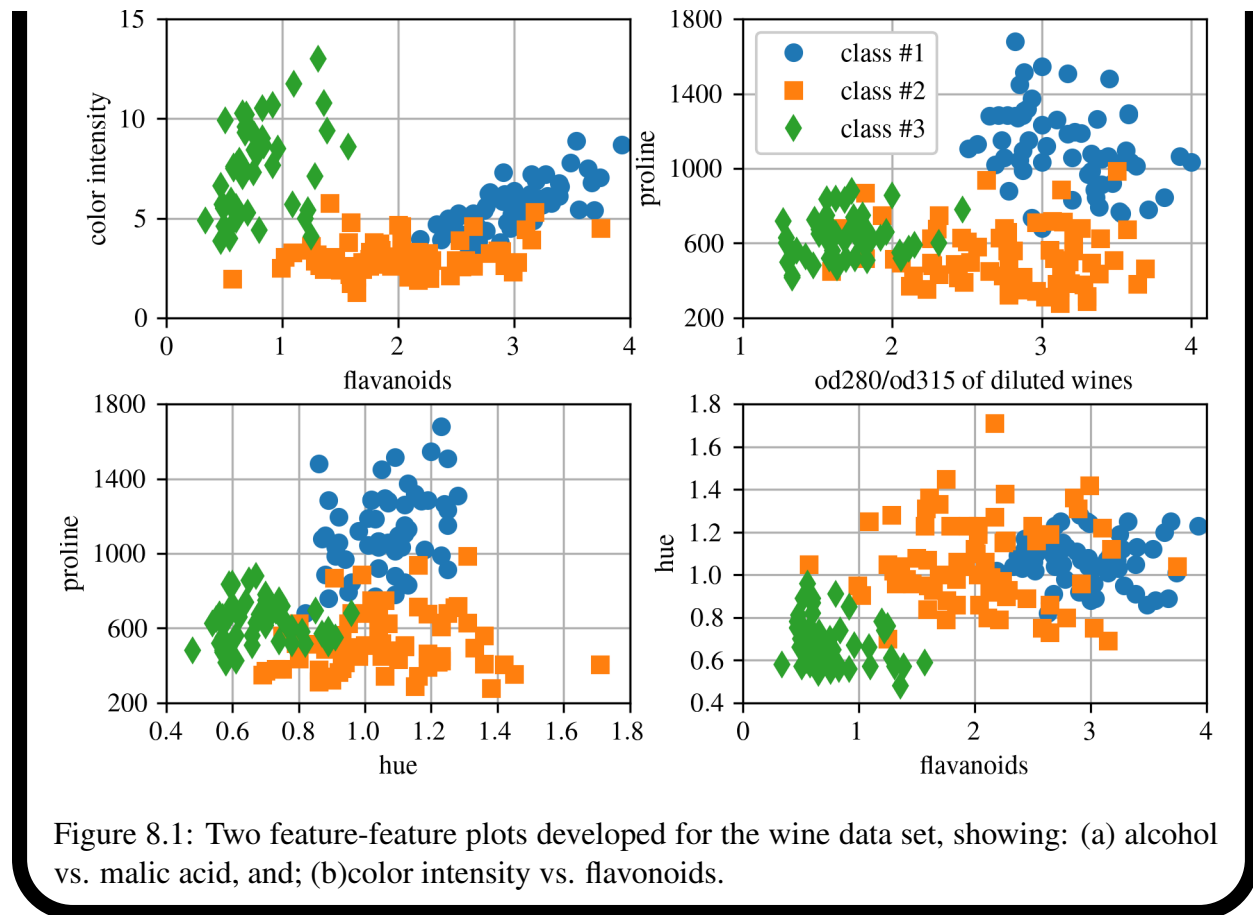
## 8 Clustering

Clustering is the process of grouping observations that appear similar without relying on external labels. In contrast to classification, where each sample is assigned to a known class, clustering is an unsupervised approach because the algorithm must identify structure in the data on its own. Figure ?? illustrates this distinction: the left panel shows a labeled decision boundary learned from annotated data, while the right panel shows the same observations grouped by similarity with no labels provided. Clustering is useful when you

1. do not have ground-truth labels,
2. suspect that hidden groups or sub-populations exist in the data, or
3. want an exploratory summary before developing a supervised model.

### **Review 8.1 Wine Dataset**

The wine dataset contains 178 samples of Italian red wine from three cultivars grown in the same region. Each sample is represented by 13 continuous measurements, such as alcohol content, flavonoid concentration, magnesium level, and color intensity. These measurements capture chemical and physical differences among the wines. Since the three cultivars form natural but partially overlapping groups, the wine dataset is commonly used as a benchmark for clustering and other multivariate analysis methods.



## 8.1 K-Means Clustering

K-means clustering is one of the most common clustering algorithms. It partitions data into a user-specified number of clusters by assigning each observation to the nearest cluster center.

### 8.1.1 Objective Function

The goal of K-means is to minimize the total squared distance between each observation and the center of its assigned cluster. This objective encourages compact clusters whose points are close to their centroids.

### 8.1.2 Assignment and Update Steps

K-means alternates between two steps. First, each observation is assigned to the nearest centroid; then, each centroid is updated as the mean of the observations assigned to that cluster.

### 8.1.3 Choosing the Number of Clusters

The number of clusters,  $k$ , must be selected before running K-means. Common approaches include using an elbow plot, silhouette score, or prior knowledge about the system.

### 8.1.4 Limitations of K-Means

K-means works best when clusters are compact, roughly spherical, and similar in size. It can perform poorly when clusters have irregular shapes, different densities, or when the initial centroid locations are unfavorable.

## 8.2 Hierarchical Clustering

Hierarchical clustering builds a nested sequence of clusters rather than requiring one fixed partition at the start. This makes it useful for exploring structure at multiple levels of detail.

### 8.2.1 Dendrograms and Linkage

A dendrogram is a tree diagram that shows how clusters merge or split as the clustering threshold changes. Linkage criteria define how the distance between two clusters is measured during this process.

## 8.3 DBSCAN

DBSCAN is a density-based clustering method that groups points located in dense regions of the feature space. Unlike K-means, it can identify irregularly shaped clusters and can label isolated points as noise.

### 8.3.1 Density, Core Points, and Noise

DBSCAN defines clusters using neighborhoods around each point. Points with enough nearby neighbors are called core points, points near a core point are border points, and points not belonging to a dense region are labeled as noise.